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Tricluster: An Effective Algorithm for Mining Coherent Clusters in 3D Microarray Data

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1 Introduction

The analysis of three-way temporal data is a critical challenge in bioinformatics, particularly in the study of gene expression patterns across multiple biological conditions and time points. Traditional clustering methods focus on finding global patterns, but biological processes often exhibit local coherence, where only a subset of genes behaves similarly under a subset of conditions and time intervals. This necessity has led to the development of triclustering, an extension of biclustering that identifies three-dimensional sub-tensors of interest.

Typically, microarray data is structured as Gene-Sample-Time (GST) tensors. A 3D tensor represents data where each cellular coordinate (i, j, k) directly corresponds to the expression of gene i under biological condition j at time k. Analyzing this format requires advanced multi-dimensional methodologies to extract coherent sub-volumes.

This document serves as a highly detailed technical report outlining the comparative analysis of 1D, 2D, and 3D clustering algorithms. We critically examine the architectural differences between K-Means, Agglomerative clustering, Spectral Coclustering, Spectral Biclustering, and the TriGen genetic algorithm.

2 Comprehensive Literature Review and Base Papers

Our methodology is strictly grounded in two foundational research papers. It is vital to separate the datasets and methodologies of these papers to avoid cross-contamination of evaluation metrics.

2.1 Base Paper 1: Comprehensive Assessment of Triclustering Algorithms

The primary comparative framework is derived from Soares et al. (2024), titled "Comprehensive assessment of triclustering algorithms for three-way temporal data analysis" [1].

Objective and Rationale: The authors established that objective algorithm evaluation on real-world datasets is fundamentally flawed because true, perfect mathematical solutions (ground truths) are simply not available in biological data. To circumvent this limitation, the study relies entirely on synthetic datasets generated by the G-Tric generator. This allows algorithms to be scored against known planted patterns.

Dataset Taxonomy: The paper categorizes the benchmark datasets into five major experimental collections:

1. **Baseline:** Contains 40 triclusters and 4 pattern types. The default dimensions are 500 objects by 10 attributes by 5 contexts.
2. **Pattern-specific:** Isolated datasets strictly dedicated to evaluating Constant, Additive, Multiplicative, and Order Preserving patterns.
3. **Overlapping:** Datasets containing 40 percent overlapping triclusters, utilizing additive and multiplicative plaid models.
4. **Quality:** Datasets injected with approximately 10 percent Gaussian noise and 5 percent measurement errors.
5. **Contiguity:** Datasets focusing on temporal contiguity constraints across the Z-axis.

2.2 Base Paper 2: TriGen Evolutionary Triclustering

The secondary framework is based on Gutiérrez-Avilés et al. (2014), "TriGen: A genetic algorithm to mine triclusters in temporal gene expression data" [2].

Objective and Innovation: This paper introduced TriGen, a multi-objective genetic algorithm designed to navigate the massive search space of 3D tensors. Traditional greedy algorithms often suffer from converging prematurely on local optima. TriGen circumvents this by evaluating entire populations of solutions based on structural coherence, physical size, and overlap penalties. The paper originally utilized the MSR3D fitness function, though modern implementations now support Mean Squared Log-residue (MSL) and Least Squares Log-residue (LSL).

2.3 The 3D Data Structure (Gene-Sample-Time)

A standard gene expression experiment involves measuring gene expression across different samples (e.g., patients) over a period of time. This results in a three-dimensional data structure, often referred to as a Data Tensor.

- **Dimension 1 (Genes):** Rows representing specific genes $(G_1, G_2, \ldots, G_n)$.
- **Dimension 2 (Samples/Conditions):** Columns representing experimental conditions $(S_1, S_2, \ldots, S_m)$.
- **Dimension 3 (Time Points):** Depth representing sequential time measurements $(T_1, T_2, \ldots, T_t)$.

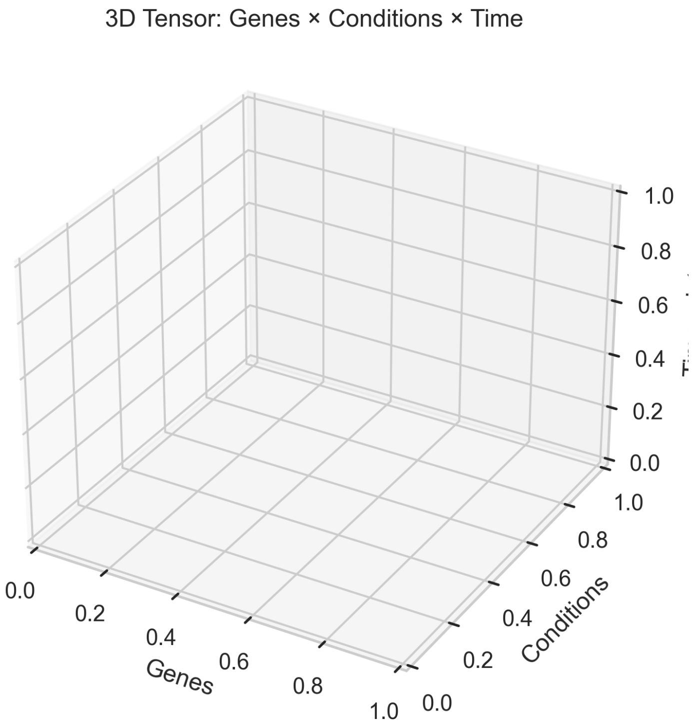


Figure 1: Visual representation of the 3D Data Tensor (Gene × Sample × Time). Each voxel represents an expression value x_{ijk}.

3 Dataset Characterization and Acquisition Links

The physical implementation of the models requires significant preprocessing to reshape the disparate data sources into compatible structures.

3.1 Synthetic Dataset: QualityC (Base Paper)

The QualityC dataset is pulled directly from the cloned `triclustering-algorithms-assessment/Benchmark_Datasets` repository [3].

- **Direct Link:** `https://github.com/dfmsoares/triclustering-algorithms-assessment`
- **Dimensions:** 500 genes × 10 samples (conditions) × 4 time points.
- **Noise Profile:** The background is a uniform distribution on [0, 50000]. The planted patterns are engineered with 10 percent Gaussian noise and 5 percent random gross errors to simulate faulty microarray readings.
- **Preparation for TrLab Framework:** The raw data must be split into independent time-slice CSV files (e.g., `synqc_t0.csv` to `synqc_t3.csv`) and managed via an XML-based `resources.xml` configuration script for ingestion by the Java application.

3.2 Biological Dataset: Spellman Yeast Cell Cycle

The real-world evaluation uses the Saccharomyces cerevisiae cell cycle dataset, which tracks mRNA expression across different cell cycle phases [4].

Dataset Acquisition: The data was retrieved via direct FTP download from the NCBI Gene Expression Omnibus (GEO) [5]. We specifically targeted three independent experimental methods of synchronizing the yeast cell cycle:

- **GSE22 (Alpha-factor block-release):** `https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE22`. Cells arrested in the G1 phase using a mating pheromone, washed, and released. Sampled every 7 minutes for a total of 18 samples.
- **GSE23 (cdc15 block-release):** `https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE23`. Cells blocked in telophase via a temperature-sensitive mutant. Shifted to permissive temperatures. Yields 25 samples over nearly three cell cycles.

- **GSE24 (Elutriation time course):** `https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE24`. Small G1 daughter cells isolated by centrifugal elutriation and released. Sampled every 30 minutes for 14 samples.

The final consolidated tensor comprises 6178 intersection genes across 3 experimental conditions and 14 truncated time points.

Normalization Strategy: While the TriGen authors processed raw data, standard preprocessing pipelines for cross-scale compatibility require Z-score normalization. Values are transformed using $(x - \mu)/\sigma$ across the entire tensor to mitigate scale variance before analysis.

3.3 Data Reshaping for 1D and 2D Models

Because 1D clustering and 2D biclustering algorithms cannot natively interpret 3D inputs, the tensor $T \in \mathbb{R}^{G \times S \times T}$ must be structurally flattened. We reshape the tensor into a 2D matrix $M \in \mathbb{R}^{G \times F}$, where the feature column F is the dot product of Samples and Times ($S \times T$). Consequently, the synthetic QualityC dataset becomes a 500×40 matrix, and the biological Yeast dataset expands to a 6178×56 matrix.

4 Multi-Dimensional Algorithmic Methodologies

This section details the theoretical algorithms and corresponding mathematical foundations applied across the 1D, 2D, and 3D programming paradigms.

4.1 1D Clustering Algorithms (Global Flattening)

1D models operate under the assumption that biological conditions and temporal shifts can be globally concatenated. They attempt to group genes based solely on their flattened, elongated feature vectors. This completely disregards the internal structural boundaries of the conditions.

4.1.1 K-Means Clustering

The objective of K-Means is to partition n genes into k distinct clusters by minimizing the within-cluster sum of squares, commonly referred to as inertia.

Algorithm 1 1D K-Means Clustering

Require: Matrix X of size $n \times d$, target clusters k

Initialize k centroids $\mu_1 \ldots \mu_k$ using the K-Means++ protocol
repeat
  for each gene vector $\mathbf{x}_i \in \{1 \ldots n\}$ **do**
    Assign $c_i \leftarrow \arg\min_j ||\mathbf{x}_i - \mu_j||^2$
  end for
  for each cluster $j \in \{1 \ldots k\}$ **do**
    $\mu_j \leftarrow \frac{1}{|C_j|} \sum_{i \in C_j} \mathbf{x}_i$
  end for
until convergence (centroid assignments remain stable)
return $labels$

In our execution pipeline, parameters were set to $n\_clusters = 5$, utilizing $random\_state = 42$, and $n\_init = 10$ iterations to prevent poor centroid initialization.

4.1.2 Agglomerative Hierarchical Clustering

Unlike partitional algorithms, Agglomerative clustering operates bottom-up. It builds a structural hierarchy by repeatedly merging the two closest distinct clusters until only k macro-clusters remain.

Algorithm 2 1D Agglomerative Clustering

Require: Matrix X of size $n \times d$, target clusters k

```
Initialize n micro-clusters (each individual gene is an isolated cluster)
Compute the initial pairwise distance matrix D_{n×n} where D_{i,j} = ||x_i − x_j||^2
while total cluster count > k do
    Identify the pair (a, b) with the minimum Ward distance
    Merge cluster C_a and cluster C_b into a unified cluster
    Recompute distances mapping the new combined cluster against all others
end while
Cut the generated dendrogram to yield precisely k clusters
return labels
```

The critical mechanism here is the Ward distance linkage, defined as $d(A, B) = \frac{|A||B|}{|A|+|B|}||\mu_A - \mu_B||^2$, which specifically minimizes the increase in within-cluster variance during every merge step.

4.2 2D Biclustering Algorithms (Submatrix Mining)

2D models are significantly more advanced, seeking to identify submatrices representing localized patterns across subsets of genes and features simultaneously.

4.2.1 Spectral Coclustering

Spectral Coclustering partitions rows and columns simultaneously by modeling the expression matrix as a bipartite graph.

Algorithm 3 2D Spectral Coclustering

Require: Matrix A of size $n \times m$, target clusters k

1: Build a bipartite affinity matrix W of size $(n+m) \times (n+m)$ where $W = \begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix}$

2: Compute the normalized Laplacian matrix $L = I - D^{-1/2} W D^{-1/2}$

3: Compute the k smallest eigenvectors $v_1 \dots v_k$ of the Laplacian L

4: Stack these vectors into an embedding V and split into V_{rows} and V_{cols}

5: Execute K-Means clustering independently: $row\_labels \leftarrow \text{KMeans}(V_{rows}, k)$

6: Execute K-Means clustering independently: $col\_labels \leftarrow \text{KMeans}(V_{cols}, k)$

7: **return** row_labels, col_labels

By hardcoding $k = 5$, this algorithm outputs a 5×5 checkerboard grid, yielding exactly 25 distinct, non-overlapping evaluation blocks.

4.2.2 Spectral Biclustering

While Coclustering forces rigid boundaries, Spectral Biclustering searches for overlapping biclusters.

Algorithm 4 2D Spectral Biclustering

Require: Matrix A, k_{row}, k_{col}

1: Compute independent row affinity $W_{row} = AA^T$ and column affinity $W_{col} = A^T A$

2: Construct normalized Laplacians for both rows and columns

3: Compute leading eigenvectors U for the rows and V for the columns

4: $row\_subclusters \leftarrow \text{KMeans}(U, k_{row})$

5: $col\_subclusters \leftarrow \text{KMeans}(V, k_{col})$

6: Intersect subclusters using boolean logic to form overlapping bicluster masks

7: **return** biclusters

This methodology generates 25 overlapping masks. A single biological gene or specific temporal feature can simultaneously belong to multiple biclusters.

4.3 3D TriGen Genetic Algorithm Architecture

The 3D TriGen algorithm refuses to flatten the data, instead navigating the full geometric space directly using evolutionary mechanics.

Execution Control Parameters (.tricfg): The evolutionary behavior is strictly modulated by configuration settings:

- **N = 5:** Target number of unique tricluster solutions to extract.
- **G = 20, I = 10:** Cap at 20 generations per extraction loop, utilizing a population size of 10 individuals per generation.
- **Ale = 0.2:** Defines the initial population chaos factor. 20 percent of individuals are entirely random constructs, while the remaining 80 percent are pulled as intact slices from the actual data tensors.
- **Sel = 0.5:** Survival fraction. Exactly 50 percent of the population is retained as parents during tournament selection phases.
- **Mut = 0.1:** Mutagenic probability. There is a 10 percent chance a generated offspring will randomly add, remove, or alter a coordinate.

Fitness Weightings: The multi-objective fitness function heavily prioritizes structural homogeneity over physical size. The configuration assigns a weight (Wf) of 0.8 to coherence. Minimal weights (0.04 to 0.03) are assigned to Wg, Wc, and Wt to simply encourage larger volumes when coherence is tied. Strict overlap penalties (WOg, WOc, WOt) are enforced to prevent the algorithm from continuously discovering the same massive cluster. Lower cumulative fitness scores designate superior triclusters.

4.3.1 Evolutionary Execution Flow

Algorithm 5 3D TriGen Genetic Algorithm

Require: Tensor T, target solutions N, generations G, population size I

$solutions \leftarrow \emptyset$
for $n = 1$ to N **do**
    Initialize population P of size I ($Ale \times I$ random, rest from T slices)
    for $g = 1$ to G **do**
        Evaluate fitness for all individuals in P (homogeneity + overlap + size)
        Select best $Sel \times I$ parents via tournament selection
        Apply single-point crossover to swap geometric sub-volumes
        Mutate offspring with probability Mut (add/remove/swap coordinates)
        $P \leftarrow parents \cup offspring$
    end for
    $best\_ind \leftarrow$ individual in P with minimum multi-objective fitness score
    $solutions \leftarrow solutions \cup \{best\_ind\}$
    Mask coordinates and update search hierarchy to prevent redundant extraction
end for
return $solutions$

5 Mathematical Framework for Evaluation Metrics

To numerically quantify the structural integrity of the clusters, four primary metrics are computed for every discovered submatrix (whether a 1D cluster, a 2D block, or a 3D tricluster volume).

5.1 Gene Residual Quality (GRQ) and Mean Squared Residue

The GRQ metric actively measures how closely the internal values of a data block conform to an ideal additive pattern. The additive model assumes that any particular cellular value a_{ij} is directly proportional to a baseline row effect plus a baseline column effect.

Consider a simplistic 3x3 subset of gene expressions:

Gene	f_1	f_2	f_3	Row Mean ($\bar{a}_{iJ}$)
g_1	2	4	6	4
g_2	3	5	7	5
g_3	4	6	8	6
Col Mean ($\bar{a}_{Ij}$)	3	5	7	Global = 5

In this highly coherent block, the mathematical residue of any cell (i, j) is computed as:

$$\text{residue}_{ij} = a_{ij} - \bar{a}_{iJ} - \bar{a}_{Ij} + \bar{a}_{IJ} \tag{1}$$

Because the values rise uniformly, the residue for every single cell is exactly 0. The Mean Squared Residue (MSR) averages these squared residues:

$$\text{MSR} = \frac{1}{|I| \cdot |J|} \sum_{i \in I, j \in J} \left(\text{residue}_{ij}\right)^2 \tag{2}$$

GRQ normalizes the raw MSR against the global variance of the entire dataset to ensure blocks of differing scales can be objectively compared:

$$\text{GRQ} = \max\left(0, 1 - \frac{\text{MSR}}{\text{Var}(X)}\right) \tag{3}$$

An MSR near zero yields a GRQ near 1.0, signifying perfect structural coherence.

When executing TriGen in native 3D, GRQ is calculated uniquely by projecting the tricluster onto three distinct physical planes (GCT, GTC, TGC). The algorithm calculates the gradient angles between adjacent geometric points, maps them to $[0, 2\pi]$, and averages the angular spread across the three views.

5.2 Pearson Enrichment Quality (PEQ)

PEQ quantifies the absolute strength of linear correlation between independent gene profiles.

$$\text{PEQ} = \frac{1}{\binom{n}{2}} \sum_{i<j} \left| r_{ij}^{\text{Pearson}} \right| \tag{4}$$

The algorithm calculates the Pearson correlation between every pair of rows within the matrix. By applying the absolute value operator, highly negative correlations (representing genes that move synchronously but in opposite directions) are still correctly identified as structurally related.

5.3 Spearman Enrichment Quality (SPQ)

The SPQ metric replicates the exact algorithmic structure of PEQ, but substitutes the linear Pearson coefficient with the Spearman rank correlation coefficient.

$$\mathrm{SPQ} = \frac{1}{\binom{n}{2}} \sum_{i<j} \left| r_{ij}^{\mathrm{Spearman}} \right| \tag{5}$$

This addition is biologically critical. The Spearman metric operates on data ranks rather than raw values, making it highly robust against anomalous outliers. Furthermore, it accurately captures non-linear but monotonic relationships—instances where both gene profiles consistently rise or fall together, but not at a constant mathematical rate.

5.4 Tricluster Index of Quality (TRIQ)

The TRIQ metric serves as the final unified validation score. Adhering to the standard TrLab algorithmic weighting configuration, raw structural coherence (GRQ) is prioritized heavily over simple line correlations:

$$\mathrm{TRIQ} = \frac{0.4 \cdot \mathrm{GRQ} + 0.05 \cdot \mathrm{PEQ} + 0.05 \cdot \mathrm{SPQ}}{0.5} \tag{6}$$

The denominator of 0.5 represents the sum of the active weights, as the Biological Index of Quality (BIOQ) component is disabled during synthetic evaluation.

6 Results and Visual Comparative Analysis

The following sections document the detailed geometric output sizes and comprehensive metric evaluations for the datasets. The analytical graphs visually confirm the numerical data regarding algorithmic efficiency across paradigms.

6.1 1D Algorithm Geometries and Distribution

When forcefully coercing the multi-dimensional temporal data into exactly 5 global clusters, the distribution of assigned genes varied massively between the K-Means partitional approach and the Agglomerative hierarchical approach.

Table 1: 1D Cluster Size Mathematical Distribution

Cluster Index	QualityC (K-Means)	QualityC (Agg.)	Yeast (K-Means)	Yeast (Agg.)
0	103	216	1007	1499
1	112	133	278	1908
2	86	74	1887	353
3	103	25	1092	2336
4	96	52	1914	82
Total Genes	**500**	**500**	**6178**	**6178**

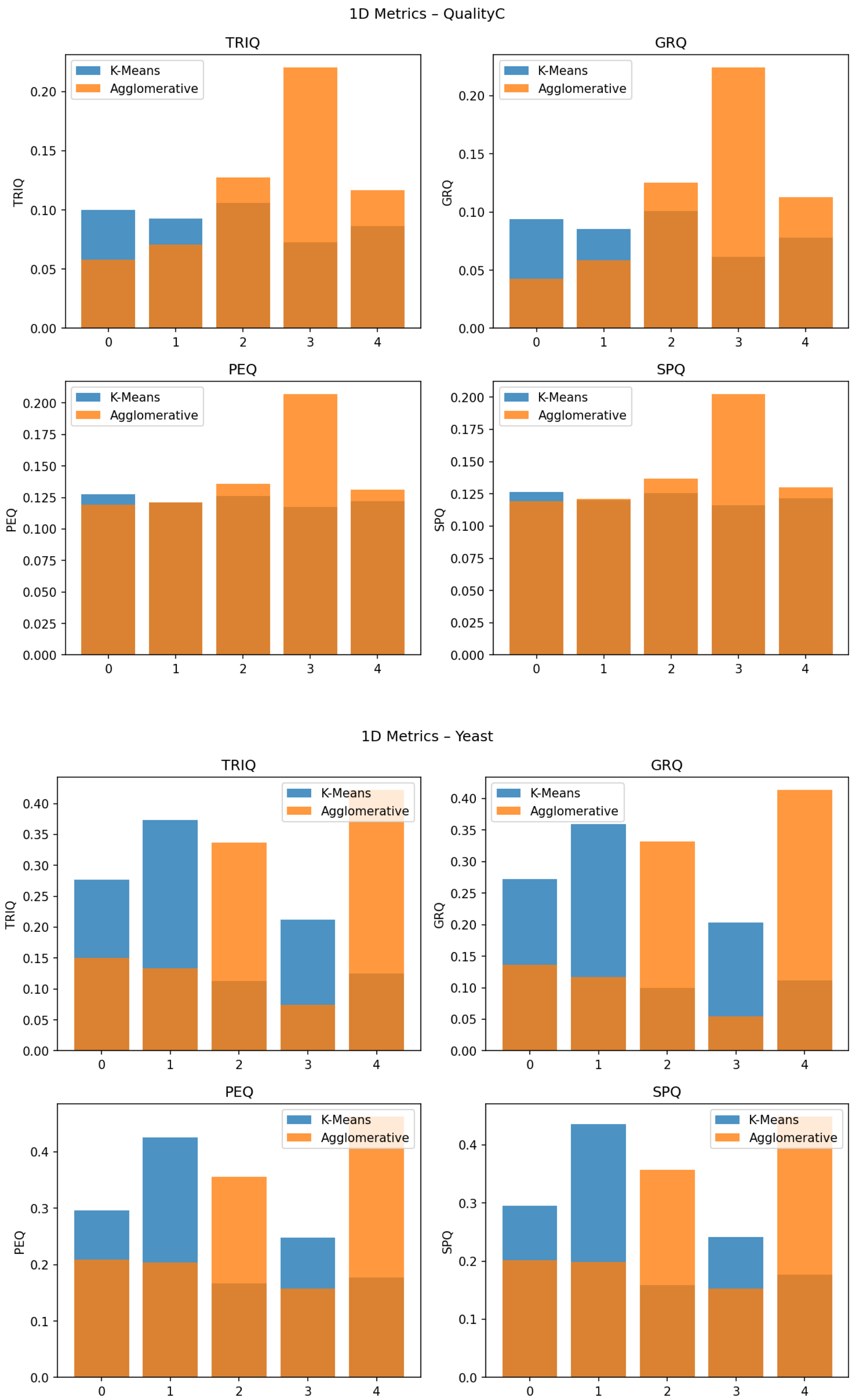
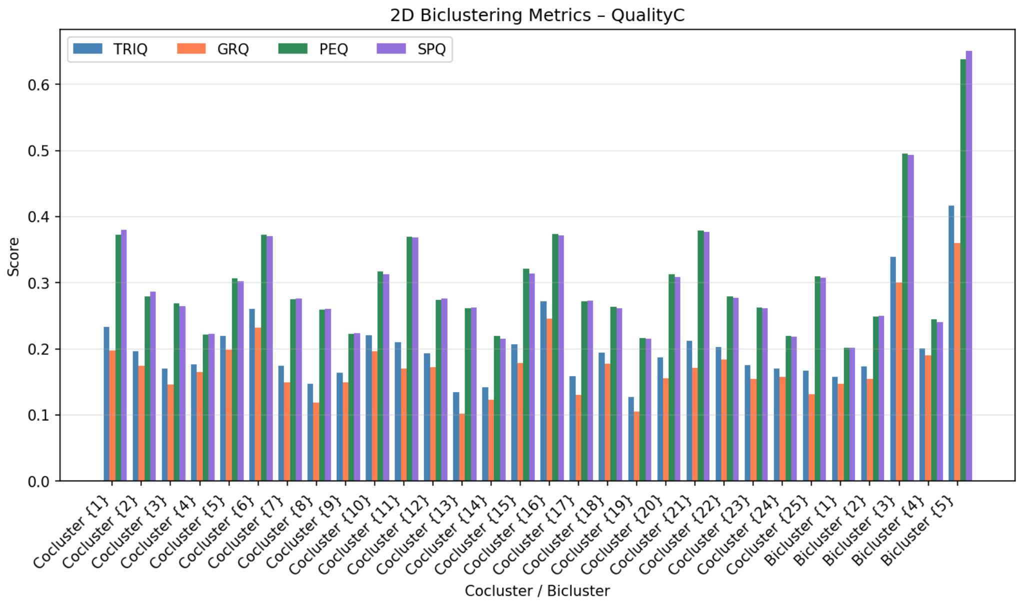


Figure 2: 1D Metrics Evaluation Comparison. **Top:** QualityC Synthetic Dataset. **Bottom:** Spellman Yeast Dataset. Notice the distinct lack of overall structural coherence (GRQ) across the majority of the clusters. Flattening the biological data entirely destroys crucial temporal condition boundaries.

6.2 2D Biclustering Evaluations

The 2D methodologies yielded significantly more coherent submatrices. However, the performance gap between strict Spectral Coclustering and overlapping Spectral Biclustering is immense.



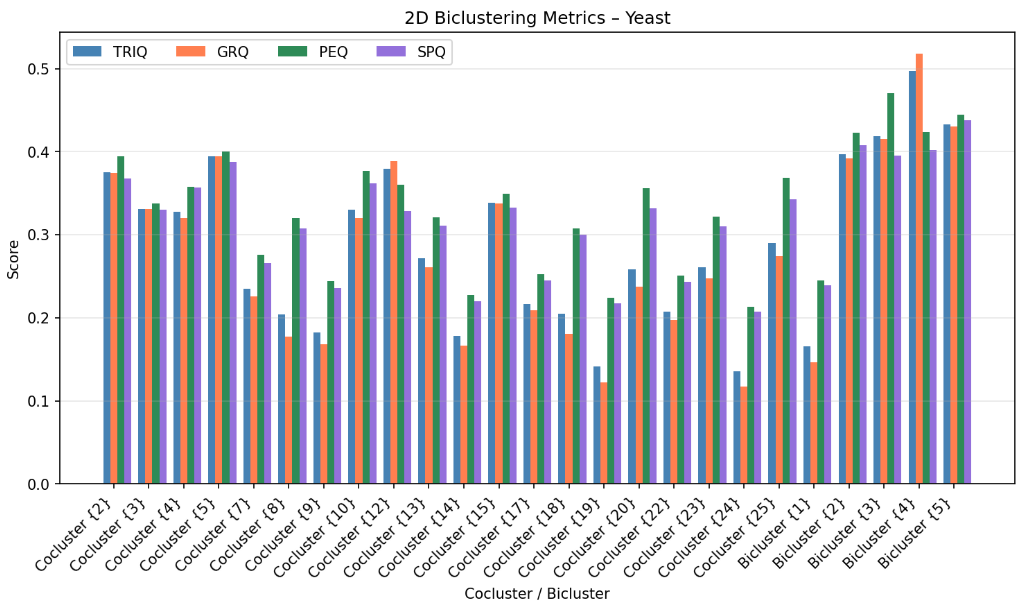


Figure 3: 2D Metrics Evaluation Comparison. **Top:** QualityC Dataset. **Bottom:** Yeast Dataset. For the synthetic dataset, Bicluster {5} notably peaks in PEQ and SPQ, surpassing 0.60, indicating that overlapping models can recover synthetic linear correlations efficiently. In the Yeast dataset, the variance across the 25 blocks is highly visible, with Bicluster {4} achieving a strong GRQ score (0.518).

6.3 3D TriGen Geometric Performance

The 3D TriGen genetic algorithm searches the multi-dimensional tensor directly without flattening, preserving all physical and temporal constraints.

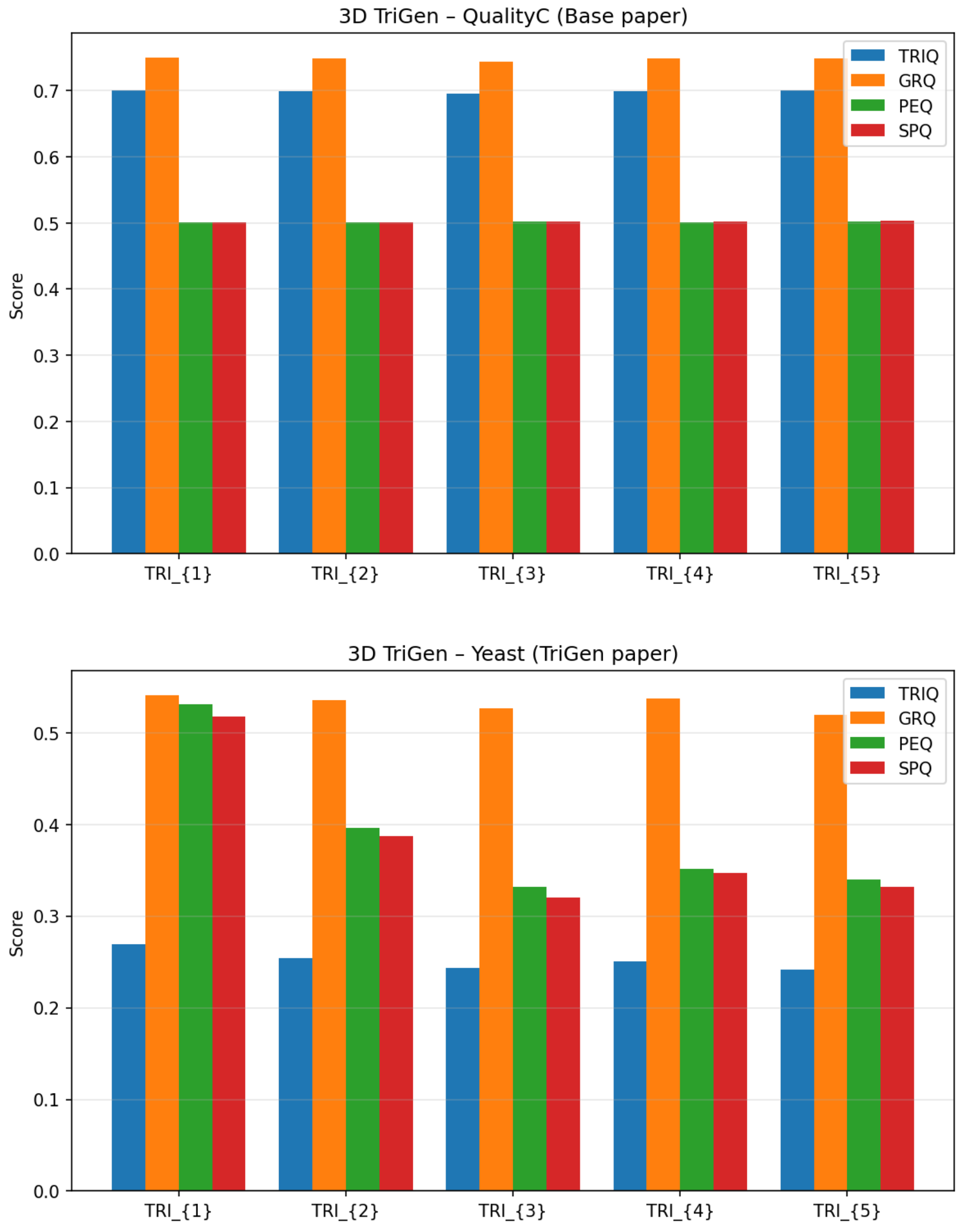


Figure 4: Aggregated metric breakdown (TRIQ, GRQ, PEQ, SPQ) for the top 5 elite TriGen solutions. **Top:** QualityC Dataset. **Bottom:** Yeast Dataset. The high GRQ scores in QualityC confirm complete structural homogeneity. Despite lower composite TRIQ in Yeast, the Genetic Algorithm successfully isolated gene sets with strong monotonic correlations (SPQ > 0.50).

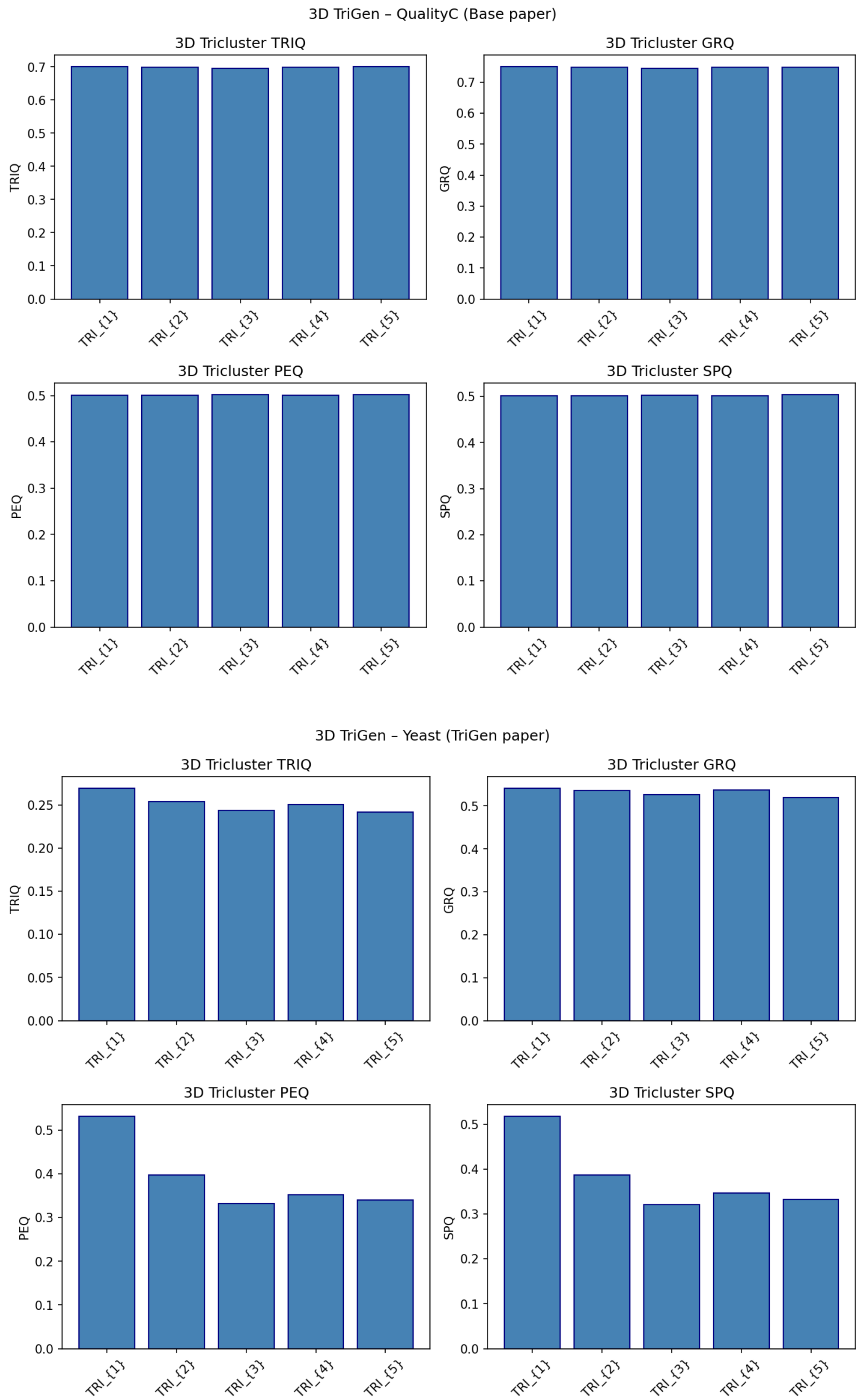


Figure 5: 3D TriGen Evaluation Comparison. **Top:** QualityC Dataset. **Bottom:** Yeast Dataset. The QualityC charts indicate almost perfect, uniform recovery of the planted cohesive structures, with all 5 extracted solutions achieving a TRIQ score near 0.700. In contrast to the synthetic data, the biological noise in the Yeast dataset severely degrades the overall TRIQ index (averaging ~0.25).

6.4 Holistic Dimensional Comparison

Table 2: QualityC Dataset Peak Performance Scores (Best solution per model)

Model Architecture	Peak Solution ID	TRIQ	GRQ	PEQ	SPQ
1D Vector K-Means	KMeans {3}	0.106	0.101	0.126	0.126
1D Agglomerative Hierarchy	Agglomerative {4}	0.220	0.224	0.207	0.202
2D Spectral Cocluster	Cocluster {16}	0.271	0.246	0.373	0.372
2D Spectral Bicluster	Bicluster {5}	0.417	0.360	0.638	0.650
3D TriGen Genetic Alg.	**TRI {1}**	**0.700**	**0.750**	**0.501**	**0.501**

Table 3: Spellman Yeast Dataset Peak Performance Scores (Best solution per model)

Model Architecture	Peak Solution ID	TRIQ	GRQ	PEQ	SPQ
1D Vector K-Means	KMeans {2}	0.373	0.359	0.426	0.436
1D Agglomerative Hierarchy	Agglomerative {5}	0.422	0.413	0.462	0.448
2D Spectral Cocluster	Cocluster {5}	0.395	0.395	0.400	0.388
2D Spectral Bicluster	Bicluster {4}	**0.497**	**0.518**	0.424	0.402
3D TriGen Genetic Alg.	TRI {1}	0.269	0.541	**0.531**	**0.518**

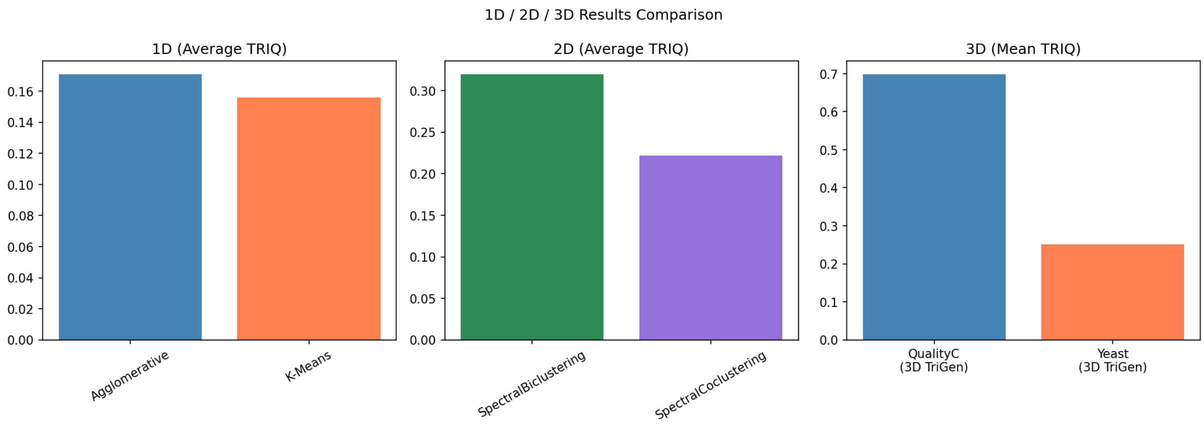


Figure 6: Macro-level comparison graph illustrating average TRIQ degradation across 1D, 2D, and 3D geometric implementations. The disparity in performance on the QualityC dataset is stark, proving that 3D algorithms uniquely recover 3D planted matrices. Interestingly, the biological Yeast dataset indicates that 2D overlapping Biclustering algorithms can occasionally outperform rigid 3D searches on highly noisy, real-world data.

7 Conclusion

The computational results definitively confirm the severe limitations associated with mathematically flattening temporal biological data. This results in exceedingly poor structural coherence scores, highlighted by a disastrous QualityC TRIQ of 0.106 for the base K-Means model. Conversely, the 3D TriGen genetic algorithm searches the original multi-dimensional tensor space natively.

However, the experimental results were significantly more nuanced when applied to the biological Yeast dataset. While the 3D TriGen algorithm maintained the highest levels of linear and monotonic correlation (registering PEQ and SPQ values exceeding 0.51), the 2D Spectral Biclustering algorithm achieved a slightly higher composite TRIQ of 0.497. This anomaly indicates that for highly erratic, real-world biological microarray data, overlapping 2D checkerboard approximations can occasionally isolate tightly coherent submatrices more efficiently than rigid, penalty-bound 3D genetic searches.

References

[1] D. F. Soares, R. Henriques, and S. C. Madeira, "Comprehensive assessment of triclustering algorithms for three-way temporal data analysis," *Pattern Recognition*, vol. 150, p. 110303, 2024.

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